Real Code Execution Flowchart & Architecture Specification

HackTheAPI // Step-by-Step Source Code Execution Across React SPA, Django REST, Docker & Ollama

Phase & Execution Step	Source Code File & Function	Real Runtime Action & Execution Logic
1. App Startup & Auth	`src/App.jsx` `src/services/api.js`	Reads JWT token from localStorage. Dispatches `axios.get('/auth/user/me')` to Django backend to populate user profile and RBAC role.
2. Login & JWT Claims	`server/apps/authentication/views.py`	Validates PBKDF2 hash in DB. Issues signed JWT access/refresh tokens containing user claims (`email, role, is_active`).
3. Lab Catalog Query	`src/pages/ModulesPage.jsx` `challenge_registry.py`	Queries `GET /api/v1/exercises/catalog`. ChallengeRegistry returns 12 OWASP labs (`ex-bola-01` to `ex-graphql-01`).
4. Docker Provisioning	`server/apps/guided_exercises/views.py`	Student clicks 'Start Lab'. Backend spawns target container session on isolated port (e.g. `Port 8102` for BFLA) and returns session token.
5. Workbench State Sync	`src/pages/ExercisePage.jsx`	Runs `useEffect()` syncing `window.__CURRENT_LAB_WORKBENCH_STATE__` with active exercise ID, OWASP type, method, path, headers, body, and output.
6. Payload Execution	`src/pages/ExercisePage.jsx`	Student clicks 'Send Request'. Axios dispatches HTTP payload directly to target container port (`http://localhost:8102/api/v1/users/carlos`).
7. Oracle Server Assertion	`server/apps/guided_exercises/oracle_engine.py`	Student clicks 'Validate Lab'. Backend `OracleEngine` asserts HTTP request payload AND queries physical container DB state (e.g. verifies user Carlos deleted). Updates `UserProgress` score (+100).
8. Screen Context Reader	`src/components/QwenFloatingAssistant.jsx`	Student submits 'guide me through this'. Component packages `window.__CURRENT_LAB_WORKBENCH_STATE__` and posts payload to `POST /api/v1/ai/chat`.
9. Local Ollama LLM Guidance	`server/apps/ai_assistant/qwen_service.py`	Django `QwenAIService` injects student progress & screen workbench state into LLM prompt. Queries local Ollama (`http://127.0.0.1:11434`) model `qwen2.5-coder`. Returns step guidance with zero cloud data leak.